

Bachelor Thesis Proposal

Robustness of Algorithmic Collusion to Information and Timing Frictions in Q-Learning Pricing

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1 Motivation and Research Question

Reinforcement-learning pricing algorithms can learn to sustain supracompetitive prices (tacit collusion) without explicit communication. The foundational result is due to Calvano et al. (2020), who showed that tabular Q-learning agents in a repeated Bertrand game consistently converge on high prices maintained by reward punishment schemes. This finding has been confirmed across various settings and has drawn significant attention from competition authorities (OECD, 2017; Assad et al., 2024).

However, the Calvano et al. (2020) baseline assumes idealized conditions: immediate observation of rivals' prices and simultaneous price setting. Real markets rarely satisfy these. Observation latency delays punishment of deviations, noise generates false signals that can trigger or mask price wars (Green and Porter, 1984), and asynchrony introduces commitment effects that complicate the interpretation of high prices (Klein, 2021; Calvano et al., 2023).

Existing work addresses these frictions only in isolation and under different baselines: Calvano et al. (2021) study imperfect monitoring in a Cournot setting with stochastic demand, Klein (2021) uses sequential pricing as the base game rather than as a friction, Asker et al. (2024) vary the learning protocol (algorithm) itself. No study combines multiple frictions within a single controlled Bertrand environment or tests whether they interact.

Main research question: How robust is learned collusion when the informational and timing assumptions of the Calvano et al. (2020) framework are relaxed individually and in combination? Specifically:

- (1) How does observation latency (delayed rival price observation) affect collusion?
- (2) How does noisy monitoring (probabilistic misobservation) affect collusion and price war dynamics?
- (3) How does asynchronous price updating affect collusion and its interpretation as genuine vs. spurious?
- (4) Do these frictions interact nonlinearly, so that individually tolerable frictions jointly break collusion?

2 Methodology

Baseline environment

I replicate the Calvano et al. (2020) setup: $n = 2$ firms, repeated Bertrand game with logit demand, a discrete price grid ($m = 15$), and independent tabular Q-learning agents with ε -greedy exploration (exponential decay). The state is the last-period price vector. Following Calvano et al. (2023), I use random exploration to ensure any collusion is genuine. The implementation is in C++, verified against baseline benchmarks.

Friction extensions

Each friction modifies only the information or timing structure, keeping the game, action set, and learning rule identical to the baseline. Setting any friction to its null value reproduces the baseline exactly (which will be my check).

(i) Observation latency ($\ell \geq 0$): Firm i observes rival j 's price from period $t - 1 - \ell$ instead of $t - 1$. This delays deviation detection and weakens punishment. I plan to sweep $\ell \in \{0, 1, 2, 3, \dots\}$ until collusion breaks down ($\ell = 0$ is baseline case from Calvano).

(ii) Noisy monitoring ($q \in [0, 1]$): With probability q , the observed rival price is perturbed to a neighboring grid point. This creates false triggers for punishment (and missed real deviations), potentially generating Green–Porter-style price wars. I will try $q \in \{0, 0.05, 0.10, 0.20, 0.30\}$.

(iii) Asynchronous updating: Only one firm updates its price per period (alternating or random selection), the other carries over its previous price. I compare synchronous vs. asynchronous modes and use off-path deviation tests (manually perturbing a firm's price post-convergence) to distinguish genuine collusion from timing artifacts.

(iv) **Interactions:** I fix a moderate latency from the latency sweep and vary noise, testing whether the collusion-breakdown threshold shifts, which would be the evidence of nonlinear compounding.

Evaluation metrics

The primary measure is the collusion index $\Delta = (\bar{\pi} - \pi^N)/(\pi^M - \pi^N)$, ranging from 0 (Nash) to 1 (monopoly). I also track convergence rates, price war frequency/duration, and off-path punishment responses.

3 Expected Outcomes and Contribution

I expect (1) collusion to degrade monotonically with latency, with a threshold beyond which it collapses, (2) noise to shift outcomes from stable collusion to collusion with recurrent price wars, eventually reaching Nash, (3) asynchrony to sustain high prices but with weaker punishment mechanisms, and (4) frictions to compound nonlinearly.

The main contribution is a controlled, like-for-like comparison of multiple frictions within a single baseline, including for the first time their interaction effects. This has policy relevance, recent work (Zhang, 2025) proposes noise injection as a regulatory tool against algorithmic collusion. My simulations can inform whether such interventions are effective and how they interact with other market imperfections.

Key References

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